

## 335 C++ Programming

### Description & Eligibility

Contestants will demonstrate knowledge and application of C++ programming, including structured design, algorithm development, and object-oriented methodologies. This event will be graded on the Windows operating system; therefore, Unix/Linux should not be used. Any secondary or postsecondary division contestant may enter this application event.

### Contestant Must Supply

- Published and/or unpublished non-electronic written reference materials
- Computer or laptop/notebook (USB-Type A or adapter required); a full-size keyboard may be used (no printer is needed) and supporting equipment (extension cords, etc.)
- Contestant must use one of these IDEs: Visual Studio 2022, Visual Studio Code, VSCode, DevC++ with Ming compiler, Code:Blocks
- Only pre-written code that is hard copied is allowed

### Contest Notes

- Method of evaluation: application
- AI Policy Level 1 *No AI Allowed*
- Carry-in and setup of equipment must be done solely by the contestant within the time allotted
- All student-produced jobs completed on-site are uploaded electronically and are not printed at the national level
- Business Professionals of America assumes no responsibility for hardware/software provided by the contestant
- No equipment, supplies or materials other than those specified for the event will be allowed in the testing area
- Previous Business Professionals of America tests and/or sample tests (handwritten, photocopied or keyed) may not be taken into the testing area. Violation of this rule will result in disqualification
- Length of event: no more than ten (10) minutes orientation, no more than ninety (90) minutes testing time, and no more than ten (10) minutes wrap-up
- Entries vary by state

### Contest Competencies

1. Demonstrate foundational understanding of computer science principles and proficiency in C++ syntax, semantics, and standard libraries to solve programming challenges
2. Design, develop, test, and maintain interactive C++ applications using modern programming practices, including control flow, logical operations, classes, file access, I/O operations, loops, methods, arrays, and advanced data structures
3. Implement efficient algorithms and utilize structured and object-oriented design techniques to create scalable and maintainable C++ software solutions
4. Create and interpret UML diagrams to model software architecture and communicate design decisions in C++ projects
5. Apply modern C++ standards to write safer, more efficient, and more expressive code using updated language features, best practices, and standard conventions
6. Use pointers to manage memory safely, prevent leaks, and ensure reliable resource handling in C++ applications
7. Utilize debugging tools within modern C++ toolchains to identify logic errors, inspect runtime behavior, and optimize program performance
8. Leverage the expanded C++ Standard Library (STL), including advanced containers, algorithms, and utilities, to write more efficient and maintainable software